

Required Readings

Required Readings for this program are open access, which means you should be able to access them at no cost. The link provided in the citation will take you directly to the reading or to a page where you can download it.

1. Stellinga, Bart et al. *Money and Debt: The Public Role of Banks*. Springer, 2021.

<https://link.springer.com/book/10.1007/978-3-030-70250-2>

- **Required pages:** Chapter 1
- **Estimated reading time:** 60 minutes

INTRODUCING STOCKS AND CRYPTOCURRENCIES

Reading Time	75 minutes
Prior Knowledge	Interest, Shorting
Keywords	Stock, Primary Market, Secondary Market, Capital Appreciation, Realized Gain/Loss, Realized, Going Public, Cryptocurrency

In the previous module, we studied fixed income securities, examining the rates and risks of borrowing and lending. In this module, we switch to equities and cryptocurrencies and study not only their expected returns but also their volatilities.

1. Choosing Stocks and Cryptocurrencies

1.1 Stocks

As a bondholder one enjoys certain privileges and one main drawback.

- A bondholder is entitled to receive a fixed flow of coupon payments at specified intervals of times. As long as the company does not default, these payments continue until maturity, when the final coupon and principal are repaid together.
- In case of a default, bondholders get paid before equity holders.
- Aside from a situation of default, the amount of the principal returned is equal to the value of the investment made.
- These payments are fixed and if the companies did very well in the market, a bondholder would not receive any gains from that upside. Indeed, this is one of the drawbacks of bonds.

A different type of security, stocks, is aimed at removing the drawback described immediately above. In fact, the ability to participate specifically in a company's upside is the strength of stocks.

Stocks are securities that represent a fractional ownership in a corporation: a **stockholder**, or **shareholder**, actually owns a fraction of the company. Stocks used to be physical certificates that entitled the holder to both a proportional amount of ownership and a share of the company's profits. Today, stock ownership is managed digitally. In the same manner, stock ownership allows the holder to vote on decisions about the management of the company. In reality, most stockholders own such a tiny portion of shares that they have no influence on the management. Collectively, the stockholders tend to vote for a board of directors. In turn, this board of directors selects a CEO and other senior managers, who in turn decide on all staffing decisions among other things.

Another name for stocks is **equities**. This is because stockholders are entitled to the equity ownership of the company. We will see more of this in a later module, but for now, think of equity as the difference between assets and liabilities. This is a simple consequence of the *Fundamental Equation of Accounting*, i.e.:

$$\text{Assets} = \text{Liabilities} + \text{Equity}$$

from which, using a simple algebraic manipulation, we easily get that

$$\text{Equity} = \text{Assets} - \text{Liabilities},$$

as claimed. From this equation, we see also that the greater the assets, or the smaller the liabilities, the greater the amount of equity the company has. This is effectively owned by the stockholders.

Liabilities and equities are means for a company to raise resources to buy assets needed to produce and/or expand their current lines of business. Companies have two ways to finance their assets: using liabilities (borrowing debts) or using equities via an issuance of shares.

Public companies sell their shares through a stock market exchange (NASDAQ, NYSE, FTSE, etc.). In specific instances, an important managerial decision may have to do with how much debt and how much equity to use to finance a company. In fact, issuing shares could be done as a way to repay back certain liabilities or even most liabilities.

Investors, on the other hand, can buy and sell shares through the stockbrokers. Stock prices are known to fluctuate during the day, but behind their decision to invest, stockholders hope that, over time, the stock they buy goes up in value. Still, this is not guaranteed, and there are instances in which investors see the value of their investment in companies go down or, albeit not too common thankfully, even be wiped out when a company files for bankruptcy or goes out of business.

Since investors hope that their investments go up in value, to achieve this goal, they have to choose good companies. Such companies would have some combination of productive assets, well-controlled liabilities, good management, and opportunities to engage in new projects or continue existing projects, creating revenue that exceeds costs. When companies have all these desirable features, they can achieve greater valuations.

The previous paragraph makes it very clear that not all investments in equities are the same, and investors try to build their portfolios combining different strategies that involve different types of companies, or equities if you prefer to use this terminology. One major approach involves selecting certain types of stocks (it is important to become familiar with these terms, as they are frequently used by those working in financial fields):

- Value stocks** or investments whose value remains stable over time. In other words, and in all sorts of market conditions, as may even slowly go up. Investors are attracted to these kind of companies because they perceive that they are possibly undervalued. Also, these companies are stable and tend to provide a reliable stream of dividends, although that cannot be taken for granted.
- Growth stocks** are instead investments that exhibit more volatility than value stocks but whose growth outpaces the market average. Investors are attracted to this class of assets as a way to earn from rapid price appreciation instead of relying on current streams of income from dividends.
- Income stocks** are stocks that offer regular and steady income, usually as dividends with low exposure to risk/volatility. In certain aspects, they are reminiscent of bonds, but indeed, investors buy them to achieve a dividend yield that is above the prevailing 10-year Treasury note coupon rate. In general, income stocks have very limited growth options, although they can earn a small return from the appreciation of the value of shares. Unlike value stocks that are desirable for the possibility of undervalued assets, income stocks are sought after as a chance to bolster fixed income portfolios.
- Dividend aristocrats** is a name that attracts attention, but they are just companies in the S&P 500 that have increased their dividends annually for the **most recent 25 years**. They are not very common since, as of May 2024 (when this lecture was created), there were only 68 stocks in this category out of about 6,000 listed on either the Nasdaq or NYSE. These stocks usually have low volatility and allow stockholders to earn a two-way price appreciation of their shares and dividends appreciation.

Once investors have decided that they want to own a company's stock, how do investors buy shares? There are two ways to buy a stock:

- Primary Market:** When the firm decides that it wants to issue new shares of stock, you can decide to buy directly at the time of issuance. Purchasing a share from the issuer directly occurs at an **initial public offering** or **IPO**. This is usually difficult to do for small investors. In fact, when a company decides to go public and issue stocks via an IPO, such an IPO is underwritten by one or more investment banks or broker-dealers. They purchase the shares from the company and then try to sell those shares to investors. Broker-dealers try to sell large pieces, and thus, these shares usually end up in the hands of large institutional investors. This is not because the small investors are not important, but the way things happen is faster and more efficient from the point of view of the bankers and broker-dealers involved.
- Secondary Market:** Once the shares of stock have been issued, they can easily change hands by trading on exchanges. If you wanted to buy existing shares, then you would have to convince an owner to sell their shares at a mutually agreed-upon price. Such convincing simply means buying at the owner's offer price on a stock exchange. Similarly, if you wanted to sell your shares, then you would have to find a buyer willing to pay a price you both agree upon. Indeed, there is an extensive amount of trading on some stocks that have been issued. From the firm's point of view, this is a good thing because the company's shares are seen as marketable and liquid (terms we will formally define later in this course).

As an investor, why do we want to purchase stock? From a corporation's perspective, why do they want to issue stock? Let's tackle that next.

1.2 Stock Benefits

As investors, from our discussion in the previous subsection, we can easily see some of the advantages that come from owning stocks:

- Capital Appreciation:** Stocks can go up in price over time. In other words, you can "*buy low, sell high*." Capital appreciation means that the asset you purchased increased in value, so when you sell it, you earn the difference in price as a profit. Therefore, you can buy stocks that you believe are underpriced or undervalued, hold on to them for some period, and sell them once they have achieved that higher value. This strategy is the essence of value trading. However, you may also buy a very popular stock that is already expensive and hold onto it while it increases even more in price. Capital appreciation simply refers to the extra value your asset achieved while it was in your possession. Anyone who's sold a home whose value has risen over time understands the beauty of capital appreciation.

- Dividend Payments:** When companies make profits, they can decide to do one of two things: first, they can pay stockholders a dividend. This dividend is income the company has made (not necessarily in the latest quarter) that can be distributed in proportion to the number of shares you own. Second, a company can decide to retain earnings they will subsequently use for funding existing or new projects. Such projects could range from building new infrastructure that will eventually lead to greater earning potential or even acquiring other companies that enhance shareholder value long term.
- Voting Rights:** As owners of the company, stockholders are endowed with rights to vote on management issues. It's somewhat of a democracy in that the more shares you own, the more votes you get. However, it's also a representative democracy in that these votes are cast for members on the board of directors. This board of directors in turn tends to vote for the executives, which in turn select and hire workers. Stockholders can have some voice but generally not over the daily operations.

- Ease of Trading:** Owning a stock means that you have an asset that is publicly traded in an accessible exchange. As a result, its price is clearly known and can be traded at or very close to that value for the several hours a day that the exchange is open. These exchanges are typically open five days a week during business hours. Therefore, it is easy to convert the stock to its cash equivalent if you decide to sell or to simply buy the stock at a fair price when you wish to own it. Throughout the day, you know how much your holdings are worth. Other assets for which no exchange exists, such as art or real estate, are more difficult to value.

- Extra Income through Securities Lending:** In the previous module, we discussed shorting. As you recall, shorting involves borrowing a security that you don't own, selling it in the marketplace (which gives you cash), and waiting for capital depreciation to occur. Capital depreciation means that the price drops, after which you would buy back, or **cover**, the short, collect the difference between the high price at which you sold and the low price at which you bought, and then return the security to its original owner. As the owner of a stock, you can be on the lending end of the security. Just like with a loan, whenever a lender is lending a valuable asset (e.g., cash in a loan), the borrower pays the lender income (e.g., interest for a loan). In the case of a stock, the borrower pays the lender the financing cost, which is a type of "daily rent" for the security. As a stock owner, if you wish to have extra income, you can sell the broker (who typically holds your security) lend out your securities to short sellers. During this time, you could still sell those shares yourself. While your long position is loaned out, you would be collecting some financing costs, in addition to the dividends that the stock pays. This extra income from financing costs continues until the borrower covers the short.

- Centralized Depositories:** The record of ownership of the stock is very safe. It would be virtually impossible for there to be a mix-up as to who owns the shares. This safety is part of a robust financial system in which the exchanges, brokerages, and real estate houses keep very careful records of who owns stock and how the stock has been traded.

Compare this to retail shares. Real estate deeds are typically held at local government (e.g., county or prefecture) levels, of which there can be thousands in a single country. In fact, it is possible for two people to make claim on the same title of land. For instance, after you pay someone for a home and receive the title, someone might come in and make a claim for the same title. This potential discrepancy explains why there is something called **title insurance**, which is insurance that if you purchase a home, you have the exclusive rights to its title. Stocks avoid this confusion by efficiently centralizing many of the processes with the exchange and clearing houses. In addition to these benefits are not specific to stocks but are really a benefit of having exchanges as the arena for trading in note to well-established clearing agencies and custodians for transferring and maintaining ownership records.

We can also look at the benefits of stocks from the point of view of the company that issues them. If a company wishes to grow, they have two primary ways of raising capital:

- Issue Bonds:** As we know from the previous module, companies that issue bonds are borrowing funds. They can raise any amount of capital that the market is willing to supply. But when that first coupon date arrives, they will have short-term cash-flow obligations. Suppose a company issued a 10-year bond. In six months, they will owe interest on the bond to each bondholder. If the company were unable to pay the interest because those funds were tied up in infrastructure for long-term projects, then they could default as early as the first interest payment. Therefore, companies issuing bonds or taking out loans have to take into consideration the timing and sizes of the obligations due.
- Issue Stocks:** Instead of borrowing funds, the company can decide to sell stocks. By issuing shares of stock, they are willing to transfer some of the ownership of the company to any members of the public willing to pay the share price for it. At an IPO, the share price is determined by an underwriting bank, which tries to place the shares with investors. Indeed, there is often a long list of willing participants eager to jump into an exciting new venture. Historically, some stocks perform exceptionally well on the day they IPO.

As seen here, the advantage from the company's point of view is that the stock never has to be paid back. It simply cedes an amount of ownership to others. Imagine if our class returned a company. Then, all the students (and the professor) would own all the assets and have rights to all future revenues. However, suppose all the collective capital were insufficient to invest in the required infrastructure. In that case, we could issue shares of stock that would dilute our percentage of ownership by granting ownership to whoever parties purchase the stock. This is what it means to "go public." For companies that have a few years of growth or require long cycles to extract a return on investment, stock issuance is much safer than bond issuance because it avoids coupon repayments. During this time, there will very likely be no dividend, but willing investors will not expect to receive them while the company is in a growth phase. This idea will be discussed later in this module.

1.3 Disadvantages of Stocks

Each of the benefits of owning stocks that we outlined can certainly be turned around. For example, as easily as capital may appreciate, it may also depreciate, which means you can get less money out of your stock then you put in. Certainly, if you had bought a house at the height of the market and then sold it after the market crashed, you would take a loss. With regards to stocks, such a loss may only occur on paper if you continue to hold the stock. However, if you actually sold the stock at a lower price, then you have **realized** the loss. So unlike other investments, like our certificate of deposit, stocks can and often do lose money.

Another disadvantage of stocks is that dividend payments may not occur. Even if a company promises to pay dividends, you might receive less than expected or nothing at all. Companies can change their strategy. First of all, a company might not have income to pay out. Second, the company may decide to retain earnings rather than distribute them to shareholders.

While holding stocks gives you voting rights, the proportion of your shares is likely to be so small that the effect is negligible on any results. Often, a majority of shares can be held by a few—even a single shareholder—who essentially control the company. Also, there can be a type of investor called an **activist investor**. For example, Bill Ackman is an activist investor. He chooses firms very methodically and buys a significant number of shares, giving him voter power, which he then uses to influence the management to "shape up or ship out." That is, improve their performance or quit. Activist investors can add volatility and uncertainty to a stock by the pressure created between shareholders and management.

In case of bankruptcy, equity holders are the last in line to be paid, and they will be paid only if there is any money left after paying the other categories of investors. Secured creditors are the first in line to be paid, possibly after the company has been liquidated. The group of secured creditors includes bondholders. Then, unsecured creditors are prioritized in a certain order: company employees, tax agencies, possible environmental claims, and, at the very end of the list, preferred and common stockholders.

Additionally, it's important to note that the market is not 24/7. Stocks do not trade 24 hours a day or 7 days a week. Suppose there is bad news when the market is closed. The company may lose value, so you want to sell your shares. However, you might not be able to do so until the next business day, which could be one, two, or even three days later, depending on weekends and observed holidays. As a result, holding stocks can be risky if there is any news that can greatly affect the stock price.

Just as there are disadvantages to the investor, there are disadvantages to the company. From the company's perspective, issuing stock means that the company can lose a lot of control over operations. When you look at companies that were founded by visionaries of their times, such as Ford Motor Company operated by Henry Ford, they have been able to keep a good share of stock in the family. As of a few years ago, the Ford family held a significant amount of the voting shares of stocks, which gave them 40% of the voting power over the board of directors. On the other hand, suppose Ford sold so many shares, say to an activist investor, that the Ford family members lose control of the company. This new activist investor can vote in a new board of directors, which can create poor leadership and management. Issuing stock can disrupt the ownership, and consequently the management, of a company. Issuing bonds avoids that problem and would have been safer in that case!

1.4 How to Value a Stock

We mentioned that one of the reasons behind investing in stocks is capital appreciation. To assess a stock's potential for capital appreciation, investors must have a reasonable estimate of the value of a stock or need to get such value from stock analysts. While quantitative analysts may not get particularly involved in this, some knowledge is nevertheless useful. There are several techniques that can be used and they are all more or less rooted in models of corporate finance. One of these techniques, the **Discounted Cash Flow model** (DCF), is very general and requires a deep knowledge of how the company works. This type of modeling is out of reach for someone who does not know the company very well. Other models like relative valuation or the **dividend discount models** (DDM) are much more accessible. In particular, the Gordon Growth Model is relatively simple yet reasonably powerful and does not require specialized knowledge of the business of a company.

1.4.1. Gordon Growth Model (GGM).

This special dividend discount model is based on the idea that the stock's present-day price is worth the sum of all its future dividends when discounted back to its present value. The price so calculated is called **fair value**. Then, if the current price of the stock is lower than its GGM value, the stock is undervalued and we should buy it. On the other hand, if the GGM value is below its current price, then we should sell the asset or maybe even short it. This model is based on the time value of money, which we are already familiar with and requires only 2 key inputs:

- Estimating dividends and dividend growth rate:** Estimating the value of a company's dividends going forward is not easy in general, but we can make certain assumptions. For instance, we could extrapolate a trend in past dividend payments and use that to estimate future dividends. In particular, we can assume that the company has a fixed growth rate of dividends and that this flow of dividends continues unchanged forever (the technical term is "in perpetuity"). So we need the expected next closest dividend per share in time, which we denote by D_1 , and the growth rate for dividends, which we call g .
- Estimating a discount factor:** Shareholders who invest money in stocks are aware that they are making a risky investment, and as with all risks, they want to be compensated for it. The firm cost of equity capital is usually taken to represent what the market demands as compensation for owning the asset and assuming the underlying risk. This cost of equity/compensation rate will be denoted by k . Note that k has to be greater than g so that the denominator remains positive.

The formula for the GGM model is the following

$$\text{Value of stocks} := P_0 = \frac{D_1}{k - g}$$

Example. Suppose that company ABC is expected to pay a dividend of \$1.5 per share, the expected dividend growth rate is assumed to be 2%, and its cost of equity capital is 8%. What is the DDM value of ABC stock? What should an investor do if the current price of ABC share is \$30?

Answer. In this case, $D_1 = 1.5$, $g = 0.02$, $k = 0.08$, and the DDM value yields

$$\text{Value of ABC stock} = \frac{1.5}{0.08 - 0.02} = 25.$$

As the DDM fair value is \$25/share and ABC is currently trading at \$30/share, the company is overvalued and the investor should sell their shares of ABC.

Example. A variation of the above problem above consists in saying that ABC just paid a dividend of \$1.4, and that its dividends are expected to grow at 2.2% a year in perpetuity. Its cost of equity is 8%. What is the DDM fair value of ABC shares?

Answer. Here, the student must pay close attention: in this case, we are not given the expected dividend (D_1), but the most dividend paid (usually denoted by D_0). We need $D_1 = 1.4 \times (1 + 2\%)$. This is not a problem because

$$D_1 = D_0 \times (1 + g) = \$1.4 \times (1 + 2\%) = \$1.43.$$

Then

$$\text{Value of ABC stock} = \frac{1.43}{0.08 - 0.02} = 24.66.$$

The GGM model is easy to use (and it requires only 3 inputs) but has some disadvantages:

- It can only be applied if a company pays dividends and this makes it suitable to value stable companies with well-established dividend cash flows. However, the method would be very unreliable or even completely inapplicable in the case of a start-up company.
- The constant dividend growth rate is also applicable mostly to mature, stable companies and difficult to justify for growth assets for which the growth rate is likely much more volatile.
- It assumes that the company will exist forever, be able to pay dividends forever, and even be able to grow them forever at a given rate.
- From a mathematical point of view, we need the cost of capital to be greater than the dividend growth rate.
- External market conditions are ignored.

1.4.2. Multistage Gordon Growth Model (mGGM).

This model maintains most of the assumptions as GGM but allows for a period of fast growth and a period of stable growth. The idea is that companies may grow fast during early stages and then they become mature and grow at a lower and more stable rate. To use this model, we need the following inputs:

- Expected dividend (D_1), to most recent dividend paid (D_0);
- Fast growth rate (g_H);
- Number of years the company remains in a high-growth period (n);
- Growth rate in the stable period (g_L); and
- Cost of equity capital (k).

Then, the formula for P_0 is more complex, but nothing too scary:

$$P_0 = \sum_{t=1}^n \frac{D_0(1+g_H)^t}{(1+k)^t} + \frac{D_0(1+g_H)^n(1+g_L)}{(1+k)^n(k-g_L)}.$$

The formula may look a little daunting, but it can easily be implemented via a spreadsheet. Note that this model can be extended from 3 to a number N of stages, if necessary.

1.4.3. Multistage Gordon Growth Model with Linearly Declining Growth Rate (H model).

This is essentially the same model as the mGGM, but it allows for the growth rate to decline linearly from high growth to stable growth over a period of H years consisting of three stages so that the assumptions are now:

- Stage 1: High growth rate is g_H ; stage 1 consists of only 1 period.
- Stage 2: The growth rate transitions from g_H to g_L over a period of H periods.
- Stage 3: From time $H + 1$ on, the company grows in perpetuity at the stable growth rate of g_L .

The formula in this case actually allows for a closed-form solution:

$$P_0 = \frac{D_0(1+g_L)}{k-g_L} + \frac{D_0(H/2)(g_H-g_L)}{k-g_L} = \frac{D_0(1+g_L)}{k-g_L} + \frac{D_0(H/2)(g_H-g_L)}{k-g_L} = \frac{D_0}{k-g_L} \times [(1+g_L) + (H/2)(g_H-g_L)].$$

Example. Suppose we are given the following information about company STU:

- current stock price = \$34
- cost of equity = 9.5%
- last dividend paid \$1.85
- growth rate (stage 1) = 7%
- growth rate (stage 3) = 3%
- years (stage 2) of linear decline = 10

What is the fair value of company STU? Is it overvalued, undervalued or fairly valued?

Answer. Using the following [spreadsheet](#), we find $P_0 = 35.01$. Hence, the shares of STU \$ are mildly undervalued.

1.4.4. True Multistage H Model.

The previous model was important to illustrate the technicalities of the H model. Now, we extend it to the following:

- Stage 1: High growth rate is g_H ; stage 1 consists of m periods.
- Stage 2: The growth rate transitions from g_H to g_L over a period of $H = n - m$ periods.
- Stage 3: From time $n + 1$ on, the company grows in perpetuity at the stable growth rate of g_L .

We could provide a formula for this case too, but it is going to become cumbersome. Therefore, we will utilize a spreadsheet and show how to apply it via an example.

Example. Company PQR is currently trading at \$27.14 per share and just paid a dividend of \$0.55. We also believe that PQR can grow at 10% for the next 5 years, and then, its growth rate will decline to 5% over a period of 10 years. Using the CAPM model, we also determine that its required return on equity is 8.5%. What is the fair value per share of company PQR?

Answer. This is contained in the following [spreadsheet](#) and we can see that the value of PQR is \$24.75 per share; in other words, the current market value of PQR is overvalued by about 9.6%. You can see from the spreadsheet how the previous model can be used to work for a true multistage H model.

1.5 Stocks vs. Bonds

Given how important these topics are, let's turn to stocks and bonds and further compare them. When a corporation issues a bond, the bond buyer is effectively giving a loan to the company with the payment date of the maturity date. During the life of the bond, the bondholder receives periodic interest payments. When a corporation issues stock, the stock buyer is effectively giving a loan to a company as well but with no repayment date. That is, the money collected when stocks are issued is never paid back by the company to the stockholder. Instead, the stock buyer is effectively getting ownership of the company directly.

That is the main conceptual difference between the bond buyer and the stock buyer. While both types of investments are providing funding to a company, the bond buyer is doing so conservatively. Their funds will almost always be paid back, so long as the company has sufficient assets. For the stockholder, there is no guarantee that they will receive anything. Typically, the stock price will not go to zero, but it can be much, much lower than when it was traded. For this reason, bonds are safer than stocks. This explains why bondholders are paid first and paid fixed amounts at specified times, all before any money can be distributed to stockholders.

For the bond buyer to receive interest and principal, the company merely needs to remain solvent. For a stockholder to receive dividends, the company must remain solvent, be profitable, and decide to pay dividends (as opposed to retaining the earnings). Since that is more difficult and uncertain, stocks tend to be more volatile. This causes stock prices to fluctuate more wildly than bond prices do.

Finally, for a stockholder to receive capital gains, the company has to create the perception (even if it is not real) that there is greater value, which justifies a higher share price. Of course, companies might fail to do this, which creates more risk. The lack of guarantees on dividends or price gains makes stocks riskier.

Example. Suppose Cindy bought 100 shares of company ABC at a price of \$12.44 each. During the 1-year dividend, Cindy decides to hold the stock, ABC pays 4 dividends of \$0.60 each. Then, immediately after receiving the fourth dividend, Cindy decides to sell the 100 shares and she is able to do so at \$15.13 per share. Compute the rate of return for Cindy. For simplicity, assume that Cindy is not re-investing the dividends as she receives them. Let's say she spends them immediately.

Answer. Let's use a scheme that shows the dates and cash flow:

- time $T = 0$: Cindy has a negative cash dividend of $100 \times \$12.44 = \1244 to buy the 100 shares of company ABC.
- time $T = 1/4$, she receives the first dividend, i.e., $100 \times \$0.60 = \60 .
- time $T = 1/2$, she receives a second dividend of \$60.
- time $T = 3/4$, she receives another dividend of \$60.
- time $T = 1$, she receives the fourth dividend of \$60 and she also receives $100 \times \$15.13 = \1513 from the sale of her 100 shares of ABC (we ignore commission and fees).

Computing the return over the 1-year period is very simple, and it amounts to

$$\frac{60 \times 4 + 1513 - 1244}{1244} = \frac{509}{1244} = 40.92\%.$$

a sizable return prompted by a generous dividend yield of $0.60 \times 4 / 12.44 = 19.29\%$, and a respectable capital appreciation of $1513 / 1244 - 1 = 21.62\%$.

2. Cryptocurrency

Many students who take this course have only a vague notion of cryptocurrencies. Hence, a quick introduction is beneficial. Cryptocurrencies are based on **blockchain technology**. As a technique, blockchain goes back to the early 1990s, more precisely to the works of Haber and Stornetta in 1991. Then, in 2002, Bak introduced the idea of **proof of work**, and finally, in 2008, Satoshi Nakamoto used these ideas to introduce Bitcoin.

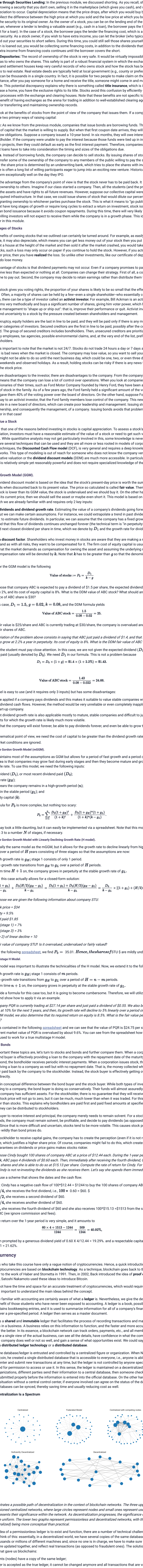
We surely do not have the time and space for an accurate treatment of cryptocurrencies, which would require hundreds of pages, but it is important to understand the main ideas behind the concept.

Those who are familiar with accounting are certainly aware of what a **ledger** is. Nevertheless, we give the definition here as well for the benefit of those students who have never been exposed to accounting. A ledger is a book, possibly a digital book, that contains bookkeeping entries, and it is used to summarize information for all of a company's financial transactions over a pre-specified period. A ledger then serves as a master document.

A blockchain is a **shared and immutable ledger** that facilitates the process of recording transactions and makes it possible to track assets in a business. A business relies on this information to function, and the faster and more accurate the information is, the better. In its essence, a blockchain network can track orders, payments, etc., and all members within the company share a single view of the actual business, can see all the details, have confidence in what the company is doing, know what the company does well or not so well, and gain a sense of what opportunities exist. We could say that a blockchain is a **distributed ledger technology** or a **distributed database**.

Traditionally, the database/ledger is entrusted and controlled by a centralized figure or organization. When Nakamoto introduced Bitcoin, he created a single distributed database that is accessible to everyone, i.e., anyone is able to see the balances and enter and submit new transactions at any time, but the ledger is not controlled by anyone specifically and there is no need for permission to access or use it. In this sense, the ledger is maintained on a decentralized basis. In centralized organizations, different parties send their information to a central database, then someone checks that everything is submitted properly before the information is entered into the official database. On the other hand, in a decentralized situation without a central control center, if everyone involved can agree on the status of the database at any time, all the decentralizations can be synchronized, thereby saving time and usually reducing cost as well.

Figure 1: Decentralization Is a Spectrum



The figure illustrates a possible path of decentralization in the context of blockchain networks. The three upper graphs depict permissionless decentralized networks, where large circles represent nodes and small ones represent users. The size of the nodes represents their significance within the network. As decentralization progresses, the significance of each node becomes more uniform. The lower two graphs represent permissionless and decentralized networks, with the lower right graph (Decentralized) being more conceptual than practical.

Clearly, if the idea of a permissionless ledger is to exist and function, there are a number of technical challenges that need to be solved. Think of this: essentially, in a decentralized world, we have several copies of the same database floating around on thousands or millions of different machines, and since no one is in charge, we have to make sure that all copies are identical, are updated together, and reflect real transactions (as opposed to fraudulent ones). The solution to these problems is what gave us blockchains:

- All participants (nodes) have a copy of the same ledger;
- Once a ledger is accepted as the true ledger, it cannot be changed anymore and all transactions that are validated are irreversible.
- Transactions must be time-stamped to avoid the possibility of adding the same transaction several times;
- All nodes agree on the validity of each block and the blockchain as the whole of the individual blocks;
- The participants remain anonymous;
- All transactions/records are encrypted; and
- Blockchains are programmable. It is a difficult task to understand without any previous knowledge of the field, but it is a necessary feature to be able to use smart contracts.

Is a blockchain secure? The question is legitimate because, without trust, people would never share valuable information with people they do not know and/or do not trust. Blockchains store data using sophisticated math and complex innovative software engines that are difficult for an attacker to create and modify. However, this does

Required Readings

Required Readings for this program are open access, which means you should be able to access them at no cost. The link provided in the citation will take you directly to the reading or to a page where you can download it.

1. Kjærland, Frode et al. "An Analysis of Bitcoin's Price Dynamics." *Journal of Risk and Financial Management*, vol. 11, no. 4, 2018, 1–18. <https://www.mdpi.com/1911-8074/11/4/63/htm>

- **Required pages:** Section 2 (pp. 2–3)
- **Estimated reading time:** 15 minutes

2. Levenkron, Hannah. *There's More than Meets the Eye: Looking at Art as an Alternative Investment*. 2011. Emory University, Honors Thesis. <https://etd.library.emory.edu/concern/etds/z316q178t?locale=en>

- **Required pages:** Abstract only
- **Estimated reading time:** 5–10 minutes

NOTE: Content presented in the Required Readings and Lesson Notes will be tested in all assignments, including Quizzes.

TYPES OF STOCKS AND CRYPTOCURRENCIES

Reading Time	60 minutes
Prior Knowledge	Interest, Shorting, Financing
Keywords	Capitalization, Dilution, Private equity, Price dynamics, Alternative investments, Commodities, Art (as an investment)

In the previous lesson, we laid the foundation for what equities and cryptocurrencies are and how their expected return and volatility can be measured. In this lesson, we'll examine different ways of categorizing these securities, as well as comparing these securities to each other and to alternative investments.

1. Large Cap vs. Small Cap

To understand capitalization, first, let's understand what a company is worth. This very simple equation is based on the market capitalization of a firm. The equation is:

$$\text{Market capitalization} = \text{Number of Shares outstanding} \times \text{Current Share Price.}$$

The right-hand side consists of two terms. The first is the number of shares the company has outstanding. Typically, these shares are created at the time of the IPO. However, a company may issue additional shares. When shares are repurchased by the company itself, these are known as **treasury stock**. In total, all of the shares combined should represent 100% of the ownership of the firm. Be sure you don't confuse these with other types of securities, such as bonds or options, which do not pertain to ownership. Stock ownership represents an exclusive and complete ownership of the firm.

Typically, the number of shares does not change often over time, but occasionally, it does. Suppose that company A decides to buy company B. Company A offers each shareholder of B 1.2 shares of A for each share they own, if they accept the merger. This requires that company A issues a certain number of new shares or uses a combination of treasury stocks and new shares.

When new shares are issued, this dilutes existing investors. For example, if a firm had issued a million shares and decides to issue a million more, then they would be **diluting** the original investors, watering down the amount of ownership those owners have. For this reason, when a company needs to raise additional funds and issue new shares, the existing shareholders typically have the right to maintain their proportion of ownership (antidilution provisions) but this has a cost. However, a discussion about dilution is beyond the scope of our discussion.

While the number of shares remains relatively fixed, the share price changes daily. In fact, it changes intraday. The share price is simply the prevailing price in the market. Generally, this is easy to determine because exchanges publish prices and volume information throughout the trading day. Suppose you were a big investor trying to buy a significant portion of the company (a takeover). You would need at least the number of shares times the current share price to purchase the company. Of course, the price will increase as existing stockholders recognize your ambition to buy their stock and now demand a premium for it, perhaps 30% or more. In addition, one would need to add the net debt as well.

In some instances, companies can decide to do a **share split** to make stocks more affordable (or even vice versa, in what is called a **reverse split**). Suppose a company has 1,000,000 shares at \$40 per share. This is worth \$40 million. The company decides to do a 2-for-1 split. Now, the new number of shares is 2,000,000 and the share price is \$20. The market cap remains the same, but this action helps to make the stock more affordable. For example, a round-lot order of 100 shares now costs \$2,000 instead of \$4,000. Splits have no effect on market cap, but they can help facilitate easier trading of the stock.

Now that we have defined market capitalization, we can categorize companies into 4 groups:

1. Large cap: \$10 billion and up
2. Mid-cap: \$2 billion through \$10 billion
3. Small cap: \$0.3 billion thru \$2 billion
4. Micro-cap: up to \$300 million (\$0.3 billion)

Let's focus on large caps versus small caps. Large caps are clearly more established companies. Sometimes, these are known as blue chips given their large size. (The term blue chip comes from the most expensive chip in poker—blue chips). These companies tend to be the ones that have been around longer and that do well in good economies as well as poor ones. What are some examples of large cap companies? These include many companies that are household names throughout the world: Microsoft, Apple, Amazon, Alphabet, and Meta (formerly Facebook) to name a few in technology, as well as older ones such as Visa, Mastercard, and Johnson & Johnson. In 2018, Apple was the first company to ever achieve a market capitalization of \$1 trillion.

Small cap companies, on the other hand, tend to be relatively newer companies. While there may be companies that have been around for years, most companies will grow to over \$2 billion dollars, especially if they are global brands. When Apple launched its IPO in 1980, its market capitalization was \$1.8 billion. (At the time, it would have been considered mid-cap, as the threshold was lower in 1980 than it is today). Market capitalization is a very clear way to distinguish these companies. Indeed, when shopping for stocks, one way is to determine what percent goes in each category of large cap and small cap. Due to the large size, large cap stocks are thought to be considered relatively safer than small-cap stocks, given that their size often comes with experienced management, existing revenue lines, and market share.

2. Growth Stocks vs. Income Stocks

We have already seen the distinction between value/income stocks and growth stocks. Growth stocks are usually those of companies that are in a phase whereby any income the company makes is used as retained earnings to grow the firm's competitiveness. This growth could be in existing markets by improving existing products, bringing new ones to market, expanding the company's geographical reach, or perhaps even expanding into different product markets.

Companies that are in the growth category are clearly identified as such. It is unlikely that an investor who wants dividends will select growth stocks. Indeed, without a stream of dividends, it may be difficult to determine if a growth stock is properly valued. Growth stocks are meant to be long-term investments that will have big capital gains. Depending on the complexity and competitiveness in the industry in which they operate, growth companies can take many years to enter the phase where they have established a serious market share with a consistent revenue. For example, Microsoft had its IPO in 1986 but did not pay a dividend until 2003. From 1986 through 2003, Microsoft was considered a growth company. In 2003, it began paying dividends. At that point, you can consider Microsoft an income stock. As you can see, companies can change as they evolve in their industry.

An income stock is simply one that is expected to provide income in the form of dividends. These companies are clearly distinguished from growth stocks because they will have a historical dividend yield. When determining whether to buy an income stock, one looks at the recent history of dividend returns. For example, a \$200 stock paying a \$10 dividend per year would be considered to have a 5% dividend yield. Typically, dividends are paid quarterly based on the earnings as well as the earnings of peers in the industry. If a dividend stock skipped paying a dividend, the response in the market would be swift and potentially extreme: namely, a sharp decrease in the stock price and negative sentiment.

When comparing stocks to bonds, income stocks clearly look more like "fixed-income securities" (bonds). Unlike bonds though, stocks do not guarantee a dividend amount or date. It is possible that stocks may miss a dividend or pay less than expected. The word "fixed" in bonds really emphasizes the guaranteed (barring a default) payout.

3. Domestic vs. Foreign Stocks

Stocks can be domestic or foreign, depending on the relationship of the company and the investor. Stocks from companies domiciled in the same country as the investor are **domestic stocks**. Stocks from other countries are **foreign stocks**. A domestic stock is one in which the investor buys the company's stock in their own country. It is true that these companies may operate in multiple nations worldwide. The key is that domestic stocks are in the same economy and same currency in which the investor does business. It is for this reason that it can be prudent to own foreign stocks as well. The reason holding foreign stocks helps is that it achieves exposure to other economies, in other markets, and in other currencies. For example, suppose you own domestic stock of a manufacturing company. Now, manufacturing stocks drop. Your domestic stock is bound to decrease.

Suppose you hold a foreign stock of a manufacturing company. In that part of the world, perhaps manufacturing stocks do not go down. You could actually make money there. Supply and demand can be very local at regional and national levels, so it's possible to lose money in a sector in one part of the world and make money in the same sector in another part of the world. The international economy is vast and complex. Therefore, it makes sense, from a balanced point of view, to hold some domestic stock and some foreign stock.

Moreover, perhaps the exchange rates are favorable. Consider the same example where the foreign manufacturing stock remains unchanged. However, suppose the exchange rate in that country improves relative to your country. That means that the foreign currency is stronger. When you go to sell your foreign stock, you will get that currency, which you can then convert into a greater number of units of your local currency. For example, if the currency in your own country loses 10% relative to the other country but the return is the same, the exchange rate profit makes it appear that the foreign stock returns 10%. The capital appreciation is due to foreign exchange rather than equity appreciation. Nevertheless, this can be profitable. Indeed, many international portfolio managers are choosing countries not only based on predictions of equities but also based on models forecasting foreign exchange rates.

4. Private Equity vs. Public Equity

One final way we'll discuss the difference in categories is the difference between a public company and private equity. Public equity means that the ownership of the shares of the company are made available to the public. Anyone who can trade on an exchange in which that stock trades is eligible to purchase shares of stock, so anyone who has sufficient funds to pay for shares becomes an owner. Therefore, when a company is about to issue shares of stock, it is known as "*going public*". Compare that with private equity. Sometimes, a company still wants to issue shares of stock but does not wish to do so through a public IPO offering. There are different reasons why a company may choose private funding rather than public. Briefly, one advantage of using private equity is the reduced number of regulatory requirements, which are easier when dealing with private equity investors since they tend to be high-net-worth individuals or businesses. Furthermore, if the number of investors is small, then it may be easier to control the ongoing management of the company. Ultimately, there could be a subsequent offering that is made public that will allow the private equity investors an even larger return by getting in "on the ground floor" of the company.

Remember that these choices are made not just for investors but from the company's perspective. A company wishing to go public has probably already brought the company to a state where it can maximize the share value for the company's founders. For example, on the day that Mark Zuckerberg issued stock for Facebook (now Meta), his wealth increased by billions of dollars. Because those stocks are now traded on a public exchange, it was very easy to evaluate his net worth. However, some companies, such as The Carlyle Group, a global investment firm, remain private investments. These companies need not worry that an activist investor or other company could buy their shares because the shares are not offered on exchanges. Ultimately, private equity ownership gives investors control of the company without the worries of quarterly earnings, potential buyouts, and other issues that affect publicly traded companies. In the long term, both are viable ways of raising funds for the original company.

5. Bitcoin

Now that we have seen different categories of stock, let us turn to different categories of cryptocurrencies. Asking an investor if they like cryptocurrencies is like asking a music fan if they like jazz. For our music example, this question is broad because there are many different types of jazz. It is certainly possible to like some forms of jazz and not others. Similarly, it is possible to like some types of stocks (blue chips) and not others. For example, suppose you don't like penny stocks, which are valued at pennies with the potential to earn big returns but are also accompanied by high amounts of risk. These companies can make well over a thousand-percent return but can also lose 100% of the original investment.

Likewise, there are many kinds of cryptocurrencies. The oldest and most popular example remains Bitcoin. As of this writing, Bitcoin is about \$800 billion dollars or 40% of the approximately \$2 trillion asset class of cryptocurrencies. Bitcoin's biggest contender is Ethereum, the second largest cryptocurrency. In addition to these two, there are many other cryptocurrencies you can choose from, just like how there is a wide selection of stocks you can choose from. Unlike stocks, however, there aren't many classifications of cryptocurrencies, like domestic and foreign, because cryptocurrencies transcend borders. Cryptocurrencies live in the global economy, in a digital world that literally operates 24 hours a day, 7 days a week. The Bitcoin exchange never closes. One way to categorize cryptocurrencies is whether they are coins or tokens. Whereas coins are built directly into the blockchain, tokens are built on top of an existing blockchain. Bitcoin and Ethereum are examples of coins and tokens, respectively. Beyond this, there are thousands of other cryptocurrencies to choose from and trade. Certain features can also help to distinguish one cryptocurrency from another.

6. Bitcoin vs. Stocks

So, let's compare Bitcoin to stocks. Stocks derive an intrinsic value because they grant ownership to companies that provide goods and services. Cryptocurrencies, as their name suggests, are effectively a currency that has implied value through the mutual trust of a currency. To better understand cryptocurrencies in the economy, it helps to understand the role of money. The required reading *Money and Debt: The Public Role of Banks* by Stellinga et al. describes three criteria for currency and focuses on the importance of trust that must remain within the currency system. In particular, be sure to read Box 1.2 Is Bitcoin money? Cryptocurrencies do not fulfill these three criteria even though they seem to have a degree of trust. Virtual Bitcoins are nearly impossible, if not actually impossible, to counterfeit due to the extreme protocols by which a Bitcoin is created. Unfortunately, though, it is possible to steal Bitcoin by accessing digital accounts. The fact remains that many cryptocurrencies are seen as a paradox by classical investors. Read section 2 of Kjerland's "[An Analysis of Bitcoin's Price Dynamics](#)" to understand the type of price dynamic Bitcoin has relative to stocks and other asset classes. Think about the "Greater Fool Theory," which suggests that the only way to make money is for someone else to be convinced that it is worth more than you paid and give you a premium for it.

7. Bitcoin vs. Other Alternative Investments

So how does Bitcoin compare to other alternative investments? Alternative investments include those investments that are outside the mainstream of equities, fixed income, and other traditional asset classes. In fact, we've already discussed one example of an alternative investment: private equity. Private equity firms invest in different companies and hold them for long periods of time with the intention of building companies up, without worrying about the whims of missed quarterly earnings or the distraction of public stockholders. They are intended for high-net-worth individuals who can afford to not have dividend income or to even suffer capital depreciation without sacrificing their quality of life. Other examples of alternative asset classes include real estate, such as homes and undeveloped land. We will be discussing the residential and commercial real estate markets in a subsequent module.

Another example of alternative assets is commodities. Commodities are an asset class that represents different products, ranging from agricultural, financial, or natural resources. Indeed, [The Commodity Futures Trading Commission](#) defines agricultural goods (except onions due to a law that banned trading futures on onions!) and physical goods, such as precious metals like gold or aluminum; natural resources, such as energy, oil, or electricity; and financial instruments, such as a currency or interest rate. That list includes a lot of items! We'll visit commodities throughout the program. One of the interesting features of commodities is the relationship of their supply and demand, as extreme weather, geopolitical risk, overproduction, pandemics, and other disruptions can easily rattle the supply/demand dynamics and affect pricing worldwide.

Our last example of alternative assets is art. At first, it's hard to see art as an investment when you think about the bonds, stocks, and other securities whose markets dominate the financial news discussing default, management changes, earnings announcements, and product launches. Many investments provide periodic cash flows. Nevertheless, art can be an intriguing asset class for its ability to find "value artists" who become very valuable over time. Read the one-page abstract from Levenkron; you will discover some interesting findings about the performance of art as an investment and how it compares to stocks.

8. Bitcoin vs. Other Cryptocurrencies

Bitcoin is the most popular cryptocurrency; it is the one that dominates the headlines. Bitcoin moves typically make headline financial news daily. Let's take a look at the demographics of Bitcoin. According to a [survey](#) reported on by CNBC journalist and best-selling author Robert Frank, most millennial millionaires use cryptocurrencies to store their wealth. Furthermore, 83% of millennial millionaires own cryptocurrencies. Therefore, Bitcoin remains an exciting investment for many. It's as if you took stock investing and fast-forwarded the movements, like in time-lapsed photography, so that a month's worth of volatility could happen in a day. The size of Bitcoin's moves dwarfs virtually every type of stock. Combined with the fact that the Bitcoin market is open more than three times the number of hours that the stock market is and that Bitcoin represents worldwide trust in a currency with accessibility, Bitcoin trading is like a 24-hour reality show with a worldwide audience. It is estimated that mentions of Bitcoin on social media appear approximately 20 times a minute (Bulao). Other cryptocurrencies simply don't have the buzz or market share. Since Bitcoin launched 15 years ago, there have been more than 15,000 cryptocurrencies, of which about 10,000 remain active. Ironically, one of those, Dogecoin, began as a joke with the image of the founder's dog on the front, only to become a serious cryptocurrency with \$18 billion market capitalization.

One thing cryptocurrencies have in common with stocks is that there are low barriers to entry. The technology of a cryptocurrency is a variation on the blockchain technology. This is a relatively low barrier to entry and explains why there are 10,000 different cryptocurrencies that trade on exchanges. The same is true for some companies. It may not be very difficult to penetrate the "advantage" the company has.

Let's look at an example of how it is possible to encroach on a company's competitive edge. Consider Peloton, for example. Peloton was the first company to produce a bicycle that you could purchase for in-home use that had a smart tablet—a proprietary screen on which riders could attend live (or recorded) classes virtually in the comfort and convenience of their own homes. Peloton tapped into a high-end market. For those who are unable to head to a gym or health facility to take a group class, Peloton offered equipment and classes with convenience, motivation, and accountability. During the pandemic, Peloton's stock achieved new highs, largely due to the fact that people were required to socially distance and the opportunity to attend an in-person class was discouraged, difficult, or dangerous. Since Peloton's launch, there have been other entrants that produced similar apps on smartphones (like Zwift), similar bikes (like Echelon), or the combination of bike and software that have competed with Peloton's model. These new entrants provided lower upfront costs and lower subscription prices to get essentially the same workout. Potential Peloton customers sought out these competitors. As a result, Peloton slashed its price. That was not enough to maintain sales. Peloton was unable to attract new customers. As of this writing, Peloton has ceased production of their bikes. Perhaps the market is already saturated with its competitors and other products being offered. Likely, the prices will drop, and new competitors will enter the market.

With such a company, it is difficult because the competitive edge is not something that is easily protected. In equities, this competitive edge is referred to as the competitive moat. A moat is a body of water that surrounded a castle in medieval times to prevent attackers from invading the inhabitants of the castle. Likewise, today's successful large-cap companies tend to have moats that provide some protection, which new entrants cannot easily breach to enter the market. As a technological example, consider Verizon. To provide fiber optic service around different regions, there must be physical towers to set up. This is not an easy thing to do, what with the purchase of land, the implementation of the technology, and the servicing. It is certainly possible to do, but there are very high barriers to entry, including high costs, accessibility to land, and infrastructure.

We discuss these issues here because the barrier to entry in cryptocurrencies is effectively the blockchain technology. Some cryptocurrencies did very well because they were the first to enter the market, and there is no telling if that success will continue, as the expectation is that new cryptocurrencies will continue to pour into the marketplace. With thousands of cryptocurrencies, can some of those displace the market share that Bitcoin has? Time will tell.

In short, cryptocurrencies are a very difficult asset class to trade, in the sense that the volume of data, the newness of the market, and the lack of the traditional methods of valuation that apply for equities or fixed income simply make it a difficult asset class to model. How can we effectively decide if we should invest in stocks, Bitcoin, or any asset? Let's go back to statistical basics, the topic of our next section.

9. Conclusion

In this lesson, we addressed different ways of categorizing stocks and compared cryptocurrencies to each other in addition to alternative investments. In the next lesson, it's time to get quantitative!

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MEASURING THE PERFORMANCE OF STOCKS AND CRYPTOCURRENCIES

Reading Time	75 minutes
Prior Knowledge	Yield, Stocks, Cryptocurrencies
Keywords	Percent Return, Log Return, Holding Period Return, Price Range, Standard Deviation, Variance, Coefficient of Variation, Sharpe Ratio

In the previous lesson, we examined categories and comparisons of stocks and cryptocurrencies. In this lesson, we will measure how to compute the expected returns and standard deviations of returns. We will also spend time investigating whether we should use arithmetic or logarithmic returns.

1. Returns: An Introduction

The notion of return on an investment is fundamental for those who work in finance because a lot of what we do deals with returns. Returns are a measure of the performance of our investments and allow us to compare performances from different investments. Therefore, it is important to have a clear understanding of this notion and the related nuances.

1.1 Arithmetic Returns

Assume that we buy a share of company STV at time t_1 and sell it at time t_2 (obviously $t_2 > t_1$). What is the return from our investment in stock STV over the period $t_2 - t_1$?

If we call S_{t_1} and S_{t_2} the price of the stock at time t_1 and t_2 , respectively, then the simplest formula to calculate the return from owning this stock over the period $t_2 - t_1$ is the following:

$$\text{arithmetic or percent return} := r_{t_1, t_2} = \frac{S_{t_2} - S_{t_1}}{S_{t_1}} = \frac{S_{t_2}}{S_{t_1}} - 1.$$

The formula we just wrote describes the *percent stock's price change* over a pre-specified time, which, in our setup, happens to be $t_2 - t_1$. This formula is not enough because stock S could pay a dividend over the period $t_2 - t_1$. To be specific, say that company S pays a dividend at time u where $t_1 < u < t_2$. Then, if we call D_u the dividend paid at time u , the earlier formula should be changed to look like this:

$$\text{arithmetic or percent total return} := r_{t_1, t_2} = \frac{S_{t_2} + D_u - S_{t_1}}{S_{t_1}}.$$

This is not really a new result for us as we already saw it in an earlier lesson, but here, we want to stress as many nuances as possible in dealing with returns since we also want to provide a complete picture of how our investment in STV performed over the specified period of time $t_2 - t_1$.

Technically, we should also consider the fact that there are commissions and fees that should be included in the formula, which tend to be much smaller compared to capital gains and dividends, so we choose to ignore them at this point.

It is possible to compute total return as a dollar amount instead of a percentage. In this case, the formula is simply:

$$\text{dollar total return} = S_{t_2} + D_u - S_{t_1},$$

but we will mostly work with percent returns and percent total returns.

From now on, we will use the terms "total return" and "return" as equivalent, and the particular formula will tell us which is the case. In any case, when the stock pays no dividends, the two formulae are the same.

1.2 Logarithmic Returns

The second way to compute returns employs the following formula:

$$\text{logarithmic return} := \bar{r}_{t_1, t_2} = \ln\left(\frac{S_{t_2}}{S_{t_1}}\right)$$

where $\ln(\cdot)$ is the natural logarithm function.

Note the use of the tilde symbol to differentiate the way the returns are computed in the two instances. The two formulae for percent return and log return tend to yield similar numbers, but they are not the same.

The formula can be easily extended to incorporate dividends, of course:

$$\text{logarithmic return} = \ln\left(\frac{S_{t_2} + D_u}{S_{t_1}}\right).$$

Example. Suppose we bought a stock on January 2, 2024, paying \$125. On February 28, 2024, the stock is valued at \$153. Compute the arithmetic return and the logarithmic return.

Answer. This is very simple:

$$\text{arithmetic return} = 153/125 - 1 = 22.40\%;$$

and

$$\text{logarithmic return} = \ln(153/125) = 20.21\%.$$

1.3 Logarithmic Returns or Arithmetic Returns?

The question of which of the two ways to compute returns is most appropriate comes up often. To answer this question, suppose that we invest an amount of S_0 dollars into an asset and we earn the following sequence of returns at time $t_1, t_2, \dots, t_n$. In addition, we assume that $t_0 = 0$. Let's call these returns $r_1, r_2, \dots, r_n$. Just to be sure that everyone is clear about the meaning, r_1 is the arithmetic or percent return between time 0 and time t_1 , r_2 represents the percent return over the period between t_1 and t_2 , and so on.

As investors, we are interested in the **cumulative return** between time 0 and time t_n . Then:

$$\text{cumulative return} := r_{0, t_n} = \frac{S_0 \times \prod_{i=1}^n (1 + r_i)}{S_0} - 1$$

and we observe that, if r_{0, t_n} represents the percent return between time 0 and t_n , then it must be:

$$r_{0, t_n} = \frac{S_{t_n}}{S_0} - 1 = \prod_{i=1}^n (1 + r_i) - 1.$$

Essentially, we are letting the returns compound continuously, i.e., we are reinvesting any gains between the time periods considered.

On the other hand, if we removed the profit in between periods so that at the beginning of each subperiod we always have an investment of S_0 , then the cumulative return would be calculated as:

$$R_{0, t_n} = \frac{S_0 + S_0 (\sum_{i=1}^n r_i)}{S_0} - 1 = \sum_{i=1}^n r_i.$$

Note the use of R instead of r to distinguish the two situations. R_{0, t_n} is the formula for total return without compounding and it is called the formula for total returns with **rebalancing**, meaning that if the return is positive, we take out the additional amount earned and spend it on something else (even use it to make another investment), and if the return is negative, we add money to our investment to bring it back to its original level of S_0 . That is where the notion of re-balancing comes into play. Students have probably noticed the use of different symbols for total returns in the two instances as we are computing total returns differently.

You may wonder why you are being taken on this journey through cumulative returns. Be patient and you will soon find out. In any case, which type of cumulative return is the one that resembles the reality of investors most?

Let's rewrite the formula for the cumulative return with continuous compounding from above so that it is easier to discuss the implications:

$$r_{0, t_n} = \prod_{i=1}^n (1 + r_i) - 1,$$

or

$$1 + r_{0, t_n} = \prod_{i=1}^n (1 + r_i).$$

Let's take the natural logarithm on both sides of the equation, which, using simple properties of logarithms, yields

$$\ln(1 + r_{0, t_n}) = \sum_{i=1}^n \ln(1 + r_i).$$

Finally, recall that $r_i = S_i/S_{i-1} - 1$, and if we replace this formula in the one immediately above, we end up with

$$\ln(1 + r_{0, t_n}) = \sum_{i=1}^n \ln\left(\frac{S_i}{S_{i-1}}\right).$$

Furthermore, we have also

$$\ln(1 + r_{0, t_n}) = \ln\left(1 + \frac{S_{t_n}}{S_0} - 1\right) = \ln\left(\frac{S_{t_n}}{S_0}\right) = \bar{r}_{0, t_n}.$$

In other words, we have shown that the total return using continuous compounding is equal to the sum of the log returns in the subperiods, i.e.:

$$\bar{r}_{0, t_n} = \sum_{i=1}^n \bar{r}_{t_{i-1}, t_i}.$$

and this property is one of the reasons for preferring the log returns. However, since

$$\ln(1 + x) \approx x$$

when x is close to zero, when the returns are sufficiently small, the two ways to compute the returns tend to produce similar results.

Another argument for favoring log returns over percent returns is that if prices are log-normally distributed, then returns would be normally distributed. But this is something important for several models used in quantitative finance that will be explained in detail in the Derivative Pricing course.

2. Return: Time-Specific Returns

Suppose we are given daily closing prices, i.e., the time indices t represent days. We can easily calculate daily returns, and we already know how to do it. However, say that we wanted to compute weekly returns using these daily data points. We can do this several ways; for instance, we could calculate returns using the closing prices on Fridays:

$$\text{weekly return week } t+1 = \ln\left(\frac{\text{Fridays price week } t+1}{\text{Fridays price week } t}\right).$$

There is nothing special about Friday's and we could have done the same using Mondays, Tuesdays, etc. Likewise, you can do the same for monthly returns:

$$\text{monthly return month } t+1 = \ln\left(\frac{\text{price last trading day in month } t+1}{\text{price last trading day in month } t}\right).$$

Months have different numbers of days, but it is typical to use the last trading day of the month for these calculations. We can calculate quarterly and annual returns as well and, in principle, returns for any timeframe we like. For quarterly returns, we could use March 31, June 30, September 30, and December 31. Likewise, annual returns typically use the values from December 31 in one year to another.

Note that if we had monthly price data, we could not compute weekly or daily returns.

It is worth saying that most closing prices for stocks are only collected from Monday to Friday when the stocks are actively traded. On Saturdays and Sundays or on holidays, because the markets are closed, there is no data, but data providers can simply extend the data by setting Saturday and Sunday prices equal to the price on Friday. This could create some problems when working with daily data because this choice may reduce the volatility in our dataset arbitrarily. In contrast, it is less of a problem when working with weekly or monthly data. Assets like cryptocurrencies do not have this problem as they can trade 24/7. When using both closing prices for stocks and cryptocurrencies, for example, we must pay attention to the different availability of data points.

An important observation has to do with the fact that if we start with N daily data for closing prices and we want to compute daily returns, then we would be able to compute only $N - 1$ daily returns. We are not able to compute the return for the first day, as we would need an extra price before the first day in our list. Also, note that if we wanted to compute weekly returns from daily closing prices and we start with N closing prices, we would be able to compute about $N/7$ returns. To obtain reliable estimates for the parameters we need in many of the models we use in finance, we require a good number of data points, so we have to consider these aspects. It should be evident that if we go for yearly returns, we need data over a very long time. While getting a long series of yearly closing prices is not a problem, we have to consider that if we use the data for modeling, the underlying models could have changed over time.

We could compute returns over different time intervals, but it would be quite complicated to compare them. Imagine that one trader takes pride in his 3% monthly returns and that another trader parades her 20% yearly returns. Who is doing better? It is typical for funds to send their investors reports about the performance of the investors' assets with the fund. Reports can be sent quarterly, semi-annually, or annually. If we received a quarterly report saying that in the most recent quarter our account had a return of 5.5%, we would probably want to know how much that is in terms of an annual return. The need to make all returns directly comparable often drives us to annualize returns.

Example. Suppose your assets in Fund X were \$100,000 on July 1. On December 31, your assets are worth \$110,000. What is the example? Well, there's a 10% gain on your \$100,000, but it only took 6 months. To annualize the return, you would effectively multiply by a factor that uses a 12-month hold. Since there are $12/6 = 2$ (two) 6-month periods in 12 months, we simply double the 6-month return. Therefore, it is a 20% gain annualized.

The main criticism of annualized returns that are obtained via an extrapolation, like the one in our example above, is that they may not reflect actual P&L.

Example. We got lucky and bought a stock yesterday for \$2 and are able to sell it today for \$2.20. That is a return of 10% in one day!! We think we are finance gurus and go around telling everyone (who cares to listen to us) that we achieved an annualized return of 3650%. "Warren Buffett who?" However, we should recognize that making 10% in one day does not indicate that we have otherworldly abilities, and we cannot be expected to repeat the same return hundreds of times. Perhaps tomorrow, the price of our stock goes down to \$1.95, and we would have to advertise a negative return instead.

The last example makes it clear that we have to distinguish an actual annual return, which is based on points that are truly one year apart, from a return that has been annualized. As a matter of fact, certain report standards, like GIPS (Global Investment Performance Standards), forbid the use of annualized returns, or the report must clearly state that they are annualized and not actual returns. In fact, when evaluating the performance of funds, we may have to pay attention to another "trick" that is sometimes employed: reporting only enough years to show the best returns and excluding years where the fund did not do too well.

Example. Suppose that a firm that started on November 1, 2023, reports a 2-month initial return of 4% on December 31, 2023, which equates to an annualized return of 26.53% (using compound interest, the most suitable for this purpose). According to GIPS (Global Investment Performance Standards) standards, the firm is required to present the 4% as a partial year's performance. The annualized return of 26.53% must not be presented. Suppose that the firm does not follow GIPS to report its performance and presents the annualized return. What should a savvy investor do?

Answer. The best approach for a savvy investor who is aware of the difference between actual performance and on-paper performance would be to de-annualize the return that was presented. Therefore, the investor can compute the actual 2-month performance as:

$$(1 + 26.53\%)^{2/12} - 1 = 0.04 \text{ or } 4.00\%.$$

Let's consider another return related to the actual time the investment is held: the **holding period return**, which is often used by investors who simply like to understand how much they made on an investment. Suppose we put \$100,000 in an investment and then sell it at \$170,000. We made a 70% return, yet it's unclear how long it took to earn it: anywhere from seven days to many years. Holding return is not annualized; it merely represents whether a profit was made.

One final return to consider is the **net return**. This will be inclusive of all incomes and costs. Suppose we've bought a stock at \$100 plus a \$2 commission, received a \$7 dividend, and then sold it for \$115. What is the net return? The formula to use can be described as

$$\text{Net Return} = \frac{(\text{money earned from investment} - \text{money invested})}{\text{money invested}}$$

which for our investment becomes

$$\text{Net Return} = \frac{((115 + 7) - (100 + \$2))}{(100 + 2)} = 20/102 = 19.6\%$$

Pay attention to the fact that the \$2 commission is the cost to enter this investment and must be included among the outflows of cash.

In addition, notice that we have not discussed taxes here. Certainly, taxes will eat into net returns. Because this class is intended for a global audience, tax rates vary wildly from country to country and even within a country from one bracket to another. As such, no discussion of taxes will be included here, but keep in mind that taxes will only lower the actual returns from these investments.

3. Moving from Returns to Volatility

We go back focused on what we are expected to make, so we can use return and its calculations to estimate what our gain is going to be. Surely if our expected returns were negative, meaning we would expect to lose money, we would likely not want to invest much, if anything, in such a venture. Buying a lottery ticket for \$2 is an exception because the loss is very small, but the potential win, while quite improbable, would be incredible! But in general, we would not expect to invest in that security.

Since we are investing in some type of risky asset, we're not sure if we're going to achieve that return. We can use historical returns to estimate what our future return is going to be. However, this expected return is nothing more than an average. When you think about an average, you're really thinking of an expected value from a random distribution. Since the distribution of results can vary, it would help to know what the expected distance is we are likely to be from that average.

In other words, even if we don't get the expected value, are we likely to be close? We are interested in the volatility around the average. This is known as the **standard deviation**.

4. Volatility: An Introduction

Standard deviation becomes the central measure of volatility but is just one way to measure volatility. The concept of volatility is much broader. It essentially addresses the fluctuation of the series itself. There are entire books written as to why volatility exists and how it can be modeled and forecasted in financial markets. Keep in mind that markets are complex systems because there are many different participants that are operating in different economies and using different models, with different tolerances for risk, cash flow needs, tax consequences (e.g., long-term capital gains are taxed differently than short-term capital gains), etc.

Therefore, the emergent property of volatility can often defy simplistic modeling that distills volatility down to a simple standard deviation. In this program, we will continually develop and refine our understanding of volatility. For now, let's consider that at any given time, there are market participants who feel a security is ready to be sold and others who feel it is the right time to purchase—that is, what helps to facilitate trading. Consequently, market prices fluctuate.

A big influence on market volatility is news. Impactful news does not arrive equally throughout the day, will vary in its importance, and will even influence some participants to make very different decisions from others. Some of these participants may be algorithms that respond to the sentiment of headlines and frantically sell shares as a result. Others may be professional managers who make the same decisions but more slowly. Finally, there are others that pause and reflect to determine if those same securities are worth purchasing at such values. One reason that volatility can change is that financial news is not absorbed in the market equally by different participants making varied decisions on different time scales.

5. Volatility: Measures

Indeed, what appears as noise to some participants can be trading opportunities for others. Consider prices that have "jumpiness" between two levels. Throughout the trading day, there is a high price at which trading occurs, "the high," and a low price, "the low." One way to measure volatility is to simply measure this price range. Price range is our first measure of volatility. It is in the units of price.

$$\text{Daily Price Range} = \text{Daily High Price} - \text{Daily Low Price}$$

Example. Suppose Friday's stock price reached a high of \$105 and a low of \$99. Then Friday's price range is \$6. Suppose further that Monday's stock price reached a high of \$106 and a low of \$104; Monday's price range is \$2. We would agree that, using price range, Friday was more volatile than Monday was. Using price range, the volatility is measured in units of price, which is how the stock trades.

To calculate the price range, we need to know the day's high and low prices. Merely knowing closing prices would be insufficient to determine the range. Throughout the trading day, think of the high as the highest point reached—the high-water mark—and the low as the lowest point reached—the low-water mark. Price range volatility looks at the difference between those two extremes—the two water marks. A very lucky day trader would have bought at the day's low price (low-water mark) and sold at the day's high price (high-water mark), so the intraday range is the maximum amount the trader could have made day trading the stock.

Another way to look at volatility is to use returns instead of prices. If we had daily returns, then we could look at the last 20 days of returns and compute their standard deviation. That means over 21 trading days (remember you need 21 days to get 20 returns), which in the New York Stock Exchange equates to about one month of trading, you would basically be calculating the standard deviation of the daily returns for one month. Stocks that are very volatile will have a high standard deviation of return. Stocks that have a low standard deviation will not show as much fluctuation.

When we use returns to compute standard deviation, the units of that standard deviation will be in return units rather than price. This gives a different perspective of volatility than price range does. Standard deviation is the expected distance you are from the average.

Suppose the average daily returns were 0.2% and the standard deviation was 0.5%. If we agree or just assume that returns are normally distributed, then we can say that 68% of the returns are within 0.2% - 0.5% on the low end to 0.2% + 0.5% on the high end. This equates to the return being between -0.3% and 0.7% for 68% of the time.

Variance is the square of standard deviation. Sometimes, people use variance and standard deviation interchangeably. Please avoid doing so.

Variance is another way to think of volatility, but its units are squared returns. In other words, variance is the average squared deviation from the average.

The problem stems from the fact that deviations can be positive or negative. A return that is below the average is a negative deviation. A return that is larger than the average is a positive deviation. When we want to understand fluctuation or volatility, we are focused on the size of the deviation and not the sign. Therefore, we need to make the negative and positive signs disappear. The easy way of doing that is to square them. (Taking absolute values would be another way). This is exactly what variance measures: squared deviations. To get back to the correct units, distance, we need to take the square root. Stated in the other direction, standard deviation is the square root of variance. It gives the expected distance we will be from the expected value. When people discuss volatility, they are likely referring to standard deviations rather than variances, since they have the same units of return. If you had a 20% standard deviation when you square 20%, you get a 4% variance. Just be sure that when you use either one, you're conscious of which one you're utilizing.

6. Volatility Divided by Return: The Coefficient of Variation

We have discussed returns. We have discussed volatility. How can we use these to make investment decisions? Ideally, we would like to have a lot of return for as little volatility as possible. More expected return means higher profits. Less volatility means we are more likely to achieve those high profits. Lowering returns reduces our profits. Raising volatility means we increase the likelihood of being far below those profits to the point where they become losses!

So let's combine the two. First, we'll start by dividing one by the other. We'll start by dividing volatility by return. Volatility divided by return is known as the coefficient of variation. Volatility is the numerator and return is the denominator. This is a classical metric from statistics. In finance, when we compute the coefficient of variation, we are basically asking the following question: how much volatility does an investment encounter to earn a given amount of return? That is, you are effectively seeing how much volatility costs per unit of return. Think of it as an exchange rate. You have to spend x amount of volatility dollars to earn y amount of return units.

Suppose you use this ratio to compare the relative risk/return relationship of two investments, such as two stocks. Stock A has a 50% volatility per unit of return. Stock B has 25% volatility per unit of return. Which of these seems like a better investment? We've intentionally not provided the specific returns and volatilities, just this ratio. This is important because the returns of the two stocks can be very different, as well as their volatility. This ratio can be thought of as using a volatility currency to purchase a return currency. How much volatility do you have to spend to earn a given amount of return? For Stock A, you have to spend 50 volatility units. For Stock B, you only have to spend 25 volatility units. Relatively speaking, Stock B is a bargain. You do not have to spend, that is, encounter, as much volatility to earn the same amount of return. The lower the coefficient of variation, the more preferable the stock is because you don't expect to face as much volatility to earn the same amount of return that you expect for the other.

Note that this assumes you had good samples of the data, that these are representative of the future price changes, and that you did your calculations correctly. All these assumptions can be questioned and perhaps even validated.

7. Return Divided by Volatility: Sharpe Ratio

Suppose instead of dividing volatility by return, we change the order and divide return by volatility. This is the more classical way to approach the problem. In this case, now, we have return as the numerator and volatility as the denominator. Suppose we change the numerator to reflect the market premium:

$$\text{Market Premium} = \text{Stock Return} - \text{Risk-Free Rate}$$

The idea combines not only return and volatility but also the fixed-income market. The fixed-income market is important because it shows what any risk-averse investor would earn by simply being in the most conservative investment: the risk-free rate. The risk-free rate is the rate earned simply for investing in a risk-free bond. Since stocks are risky, they should earn returns in excess of that return. If a stock were to earn less than the risk-free return, then a rational investor would forgo the volatility of the stock market and stick with risk-free investments, which protect the principal while still earning some return.

The numerator in our example is the market premium. So, suppose for stock C, the market premium and standard deviation are 7% and 10%, respectively. Suppose for stock D, the market premium and standard deviation are 8% and 12%, respectively. Which is the preferred investment?

$$\text{Stock D: } = \frac{\text{Market Premium}}{\text{Volatility}} = 8/12 = 0.67 \text{ or } 67\%.$$

In this case, stock C has a higher return per unit of volatility (risk) than stock D does. We would prefer the investment that has a higher amount of return per unit of risk. Now, this is the opposite of our coefficient of variation. When we express volatility units per return, we would like the lower number. When we express return units per volatility, we prefer the larger number. Here, the larger number means we expect to earn more return per unit of volatility.

In this course, we'll name this ratio (keep a *sharp* lookout for it) and use it extensively to compare investments, but for now, the important thing is that we've seen how return and volatility can be combined to compare investments. This can just as easily work if these securities are from different asset classes. Return and risk help encapsulate the key statistical properties of a financial asset's performance. In the next lesson, we'll dive deeper into the details of these statistics.

8. Conclusion

In this lesson, we measured returns and standard deviations. These measures tell us what we expect to earn (or lose) and how volatile the actual value can be. In the next lesson, we'll try to assess if these returns come from a distribution that resembles a Gaussian or normal distribution.

MODELING THE PERFORMANCE OF STOCKS AND CRYPTOCURRENCIES

Reading Time	60 minutes
Prior Knowledge	Interest, Mean, Standard Deviation
Keywords	Normal (Gaussian) Distribution, Skewness, Kurtosis, Student's T Distribution, Lognormal Distribution

The previous lesson introduced us to the mean and standard deviation of a distribution. This lesson attempts to use those parameters to describe a data distribution that can resemble a normal distribution. If the resemblance is good (that is, if they are similar), then they should NOT have skewness or kurtosis. This lesson will define skewness and kurtosis and assess whether our data has significant amounts of either.

1. Evolution of Prices over Time

The first thing that we will consider is simply graphing the prices over time. Normally, if you were to pull up a financial stock series, you would see its price over time. That is, the y-axis would indicate the price range and the x-axis would indicate a period of time, say, the last two years. For stocks, prices never go below zero. For bonds, yield would be on the y-axis, and it is possible that yields can go negative. But for now, let us consider that price goes on the y-axis. The time series graph simply graphs the price of the security over the period of time indicated along the x-axis.

A comparison between the distribution of closing prices and log returns can be seen in the following [spreadsheet](#) that plots the histogram of daily returns for the Apple stock and which you can easily manipulate (just be sure that you change the data interval in the charts if you redefine the time interval). You should notice that the histograms seem to confirm something we said in Lesson 3 of this module: if stock prices are log-normally distributed, returns are normally distributed.

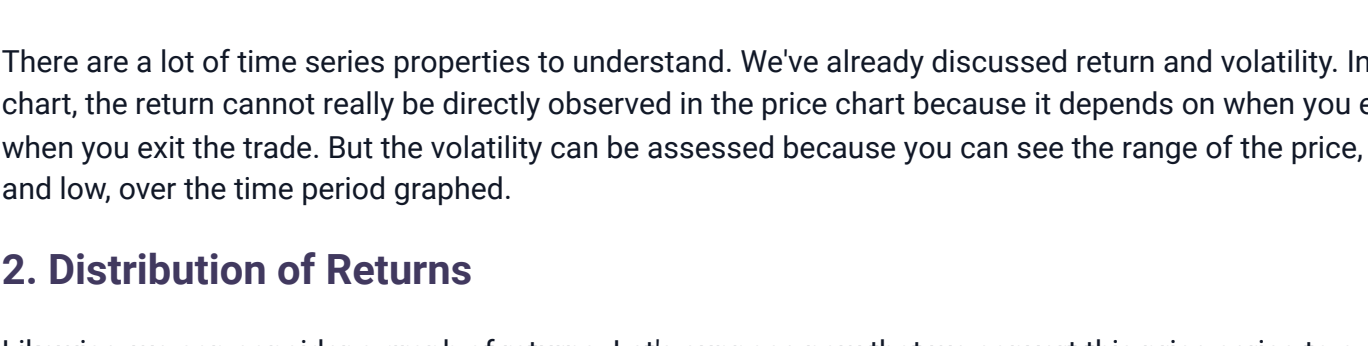
Example. *Why is it not possible for stock prices to be normally distributed?*

Answer. A normal distribution allows for negative values and prices cannot become negative.

When we are reading the plot in Figure 1, there are a lot of questions we can ask. For example,

- How high does the series go?
- How low does the series go?
- Does it oscillate between these two points frequently?
- Does the series tend to rise or fall over time, creating a trend?
- Does there tend to be more fluctuation at one time than there is at another time?
- Does the series rise more abruptly than it falls?

Figure 1: Netflix Stock Price Rallies during COVID-19



There are a lot of time series properties to understand. We've already discussed return and volatility. In the price chart, the return cannot really be directly observed in the price chart because it depends on when you enter and when you exit the trade. But the volatility can be assessed because you can see the range of the price, the high and low, over the time period graphed.

2. Distribution of Returns

Likewise, we can consider a graph of returns. Let's suppose now that we convert this price series to a daily return series. The y-axis is now in units of return. Some days are positive, and others are negative. That is why 0 is now approximately the center of the distribution of these returns. You also notice that the series is more volatile because we do not always have positive returns following positive returns or negative returns following negative returns. As such, the return chart is effectively showing what the daily returns are over time. Note that this is different from a cumulative return. For example, let's say you start trading on the first of the year. A cumulative return would show you how your trade is performing, so the other returns are effectively saying what the return is today, without a fixed moment in time.

As always, the x-axis represents time. Please pay close attention to these timescales. When you visualize financial time series, you will notice that scales can range from within minutes of a trading day to literally more than 100 years. When the time scale is very small, then you must wonder what the frequency of return is. When the time scale is very large, the timescale elongates, and oftentimes, you may be looking at quarterly or even annual returns. It is prudent to match the time horizon with the type of return. When you want to emphasize financial properties, it is appropriate to use shorter time scales. When you are trying to show larger economic cycles, it is appropriate to use monthly or quarterly returns.

Different intervals of returns will emphasize different points of the time series. For example, daily returns can be filled with noise of idiosyncratic news as well as a lot of technical details that do not really affect the long-term viability of the performance. On the other hand, monthly returns might miss some trading opportunities that can only be observed when you're looking at a daily or even intraday scale. As you progress through your coursework, pay attention to the type of data that you collect. It is always easier to extend the return interval rather than to truncate it. For example, if you had daily data, you could produce any return from daily or slower, like weekly, monthly, quarterly, etc. However, if you only had monthly data, it is not possible to construct daily returns. Therefore, it will become helpful to work with as high a frequency of the data as possible and then decide and calculate the appropriate interval of return series. For example, in future coursework, such as Econometrics, you will study properties of time series that may only exist when using daily returns but not if using monthly returns.

What both time-series plots have in common is that they are graphing either the price or return over time. That is, they are keeping the time order of data and in no way are they condensing the data into a distribution. However, it makes sense to collapse the returns to drop the observation times and focus exclusively on the returns. In doing so, we are analyzing the data not as a time series but merely as a sample. When we do this, we are focusing on the distribution of returns, and that will be the topic of the remainder of this lesson.

When we examine the distribution of returns, we are really trying to assess if the distributions are normally distributed. What is so great about normal distribution? Well, let's just say that if you have a normal distribution, then there is a lot of simplification that occurs. The normal distribution is an easily managed distribution. Imagine that you have a lot of data points: not merely thousands but millions or even billions. If they are well approximated by the normal distribution, then those data points can effectively be represented by two single values: a mean and a standard deviation. It is like a political system where the data points are voters, and they elected representatives: a mean, which represents their average or expected value, and a standard deviation, which represents a distance around where you would expect them to be around that average. Using these two parameters and the properties of the normal distribution, it is easy to estimate percentiles and probabilities from the data. This will help in many financial applications such as trading strategies, portfolio optimization, and risk management, all of which require the estimation of moments.

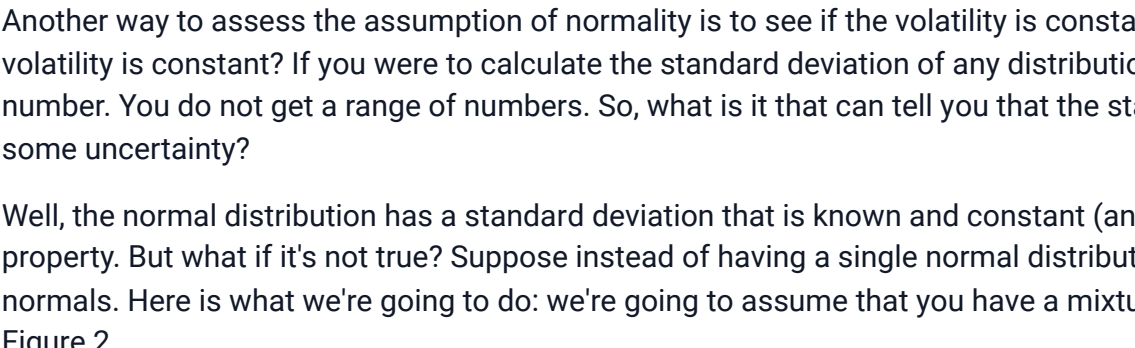
We cannot merely assume that data comes from a normal distribution. In fact, much of science does assume this because of a property known as the **Central Limit Theorem**. What does the Central Limit Theorem say? If you have random variables coming from independent and identically distributed distributions, then the sum of those variables approaches a normal (or Gaussian) distribution. Is that the case for our financial series? In other words, will economic, financial, and individual decision makers combine their results in such a way that causes the cumulative distributions of their decisions to be independent and identically distributed? Stated practically, does a stock's behavior on Monday come from the same type of random distribution that it does on Tuesday? And again on Wednesday? Thursday? Friday? Unfortunately, this is shown not to be true. In other words, because of the dynamic nature of finance, because of the ability of market participants to change their behavior, we cannot merely assume that each trading day is both independent and identically distributed to other trading days. As a result, we cannot just assume that data is normally distributed. So, what helps is to assess whether the distribution is Gaussian by checking on its basic features.

3. Are Returns Symmetric?

Is the distribution of our returns symmetric? If we assume that the returns are normally distributed, then the answer is yes. However, this happens because symmetry is built into the Gaussian distribution. Let's consider a standard normal, which is a Gaussian with a mean of 0 and a standard deviation of 1. If you were to place a mirror along the y-axis of a normal of a standard normal distribution, then everything that you'd see on one side is in fact what you see on the other. So indeed, if you just had half the distribution and the mirror, you'd have an exact match of the distribution on the other side. This is the definition of symmetry.

Some distributions are not symmetric. In general, the most typical case of lack of symmetry occurs when we can have distribution with longer tails on the left (negatively skewed) or the right (positively skewed). If we determined that the distribution of returns of a certain investment is negatively skewed, this is loaded with practical consequences. In fact, that there returns that can be from zero on the left side, that means there are potentially large losses. If we insisted in using a symmetric distribution like a normal for example, we could end up grossly underestimating the loss potential for the investment because we are ignoring the more extreme values in the left tail. Indeed, this is a common trend in finance: in equity markets, there tend to be days of panic where there are large negative drops but not corresponding large positive gains, which would represent days of great optimism. Later in the program, we will study behavioral finance and learn that humans are hard-wired to remember painful events (e.g., large stock drops) more than pleasant events (e.g., large stock gains), so we tend to avoid painful events at greater lengths.

Figure 2: The 3 types of ideal possible skewness: positive (long right tail), symmetric, and negative (long left tail)



Now, finance is not physics. It is more like a combination of evolutionary biology, cognitive biases, and socially influenced decision making. That means that financial returns cannot always be explained as if they came from physical systems. They represent the millions of actions and interactions of both humans and computers running programs, whose values are in some way still decided by humans.

The result is that if returns are no longer symmetric, then we may not be able to assume that the normal distribution is sufficient. Symmetry is an easy way to check because the absence of symmetry is readily seen in a graph.

There are two ways to do this visually:

1. Draw a histogram. Are the bars equally populated on both sides?
2. Draw a density plot. Think of dropping a mound of sand on each data point. If the aggregate shape looks like a normal distribution, then the overall distribution can be thought of as normal. Departures from normality will be easy to spot.

If we don't want to rely on simple visuals, there are several statistics to measure skewness. An easy one is that known as Pearson's first skewness coefficient and defined as

$$\text{skewness} = 3 \times \frac{\text{mean} - \text{median}}{\text{standard deviation}}.$$

It is easy to see from this formula that when the distribution is symmetric (which one can show necessarily implies that the mean and the median of the distribution are the same), then skewness is zero. When we are not given a distribution, but instead are only given a certain number of observations from a population (almost always the case in applications), $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$, then we can use the following formula:

$$\text{skewness} = \frac{\frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})^3}{\left[\frac{1}{n-1} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})^2 \right]^{3/2}}.$$

However, in this course, we are not going to do any math or use the formulae in any meaningful manner. What is important for students to remember are the notions of symmetric, positive and negative skewed, distributions and their possible implications for risk.

For a symmetric distribution, the standard deviation could be an acceptable measure of risk. For skewed distributions, in general, it is not.

4. Is Volatility Constant?

Another way to assess the assumption of normality is to see if the volatility is constant. How can one check that volatility is constant? If you were to calculate the standard deviation of any distribution, you would get a single number. You do not get a range of numbers. So, what is it that can tell you that the standard deviation itself has some uncertainty?

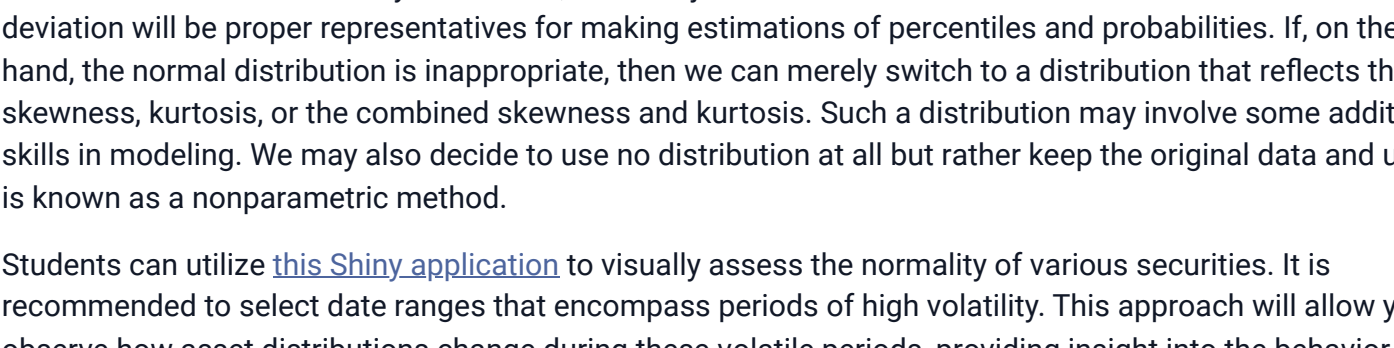
Well, the normal distribution has a standard deviation that is known and constant (and finite). This is an important property. But what if it's not true? Suppose instead of having a single normal distribution, we produce a mixture of normals. Here is what we're going to do: we're going to assume that you have a mixture of two normals. See Figure 2.

80% of the mixture is a normal distribution with a mean 0 and a standard deviation of 1.

20% of the mixture is a normal distribution with a mean 0 and a standard deviation of 4.

When you compute the standard deviation of this mixture of normals, you get a value around 2.

Figure 3: Comparison of Gaussian (Normal) and Non-Gaussian Distributions



Is this value a reliable measure of volatility?

The answer cannot be understood by merely looking at the volatility. Think what the second moment of a distribution measures: the variance. It gives the squared variation around the mean. What we'd like to know is how accurate that squared variation is. We can get an idea of this from the excess kurtosis. Think of the excess kurtosis as a quality check on the variance. For a normal distribution, the variance is known with certainty (since the standard deviation is known). Therefore, there is no uncertainty about the variance, and there is no excess kurtosis. But suppose we take an empirical distribution and find that there is excess kurtosis. Then, we can likely conclude that this distribution did not come from a normal distribution. Indeed, when we measure the excess kurtosis of the individual normal distributions, they are no different from 0. But when we measure excess kurtosis of the mixture of normal distributions, we get a value around 7. We have a non-Gaussian distribution.

5. Are Returns Normally Distributed?

The Gaussian distribution gives us proportions of how often we could be within a range. For example, 68% of the time, we expect to be either one standard deviation below or one standard deviation above the mean. Additionally, 95.5% of the time, we expect to be within two standard deviations below to two standard deviations above the mean. Then, 99.7% of the time, we expect to be three standard deviations below to within three standard deviations above the mean. So, as you can see, the normal distribution is like a powerful zip file that can condense any number of data points into a distribution that can recreate percentiles and probabilities by storing two parameters: the mean and standard deviation. But suppose we did not have a Gaussian distribution? Then that framework is no longer appropriate. The standard deviation is no longer a reliable way to determine what the expected distance is from the mean.

We need to do something that helps us recognize when this assumption of normality does not hold well. And so, we can simply measure excess kurtosis. An excess kurtosis significantly larger than 0 means that the assumption of normality is likely violated.

This means that the assumption of normality is not very well fulfilled. In other words, when we say that returns are normally distributed, but we have excess kurtosis, we are violating a basic property of the normal distribution. That property is that the standard deviation is essentially known and constant. In our example, the mixture of normals, the excess kurtosis is around 10. This high excess kurtosis indicates that there is effectively a variance to the variance. Later in this program, we will call this "the volatility of volatility" or the "vol of vol" when we price options!

Failing to get the right standard deviation can mean disaster. We not only get the wrong estimates of percentiles, but we are also fooled into having a false sense of confidence that it was correct. Relying on the normal distribution when it is not true gives us a very false sense of security!

The art of financial engineering is being able to use appropriate models for volatility for the data at hand. Keep this in mind as you progress through the program because the ability to progress in your field as a financial engineer is the ability to recognize where models work and where they break down.

6. Normal or Not?

If you wanted to introduce skewness, there are distributions that allow us to do so. We will examine these in the Financial Data course. These include:

- the F distribution,
- the skew-normal distribution,
- the log-normal distribution, and many others.

There are formal tests for determining if a distribution has skewness. There are even robust measures that can handle skewness by focusing on either ranks instead of values or even by choosing to use only one side of returns (usually the negative ones) and ignoring positive returns (e.g., semi-variance).

Another important concept to consider in the shape of the distribution is kurtosis. In its essence, the notion of kurtosis investigates whether a distribution is peaked or flat. The relevance is that a peaked distribution exhibits lighter tails, while a flat distribution tends to have heavier tails and this is relevant when we want to assess the risk of certain investment.

For a sample of observations $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ from a population, one formula to compute the sample kurtosis is the following:

$$\text{kurtosis} = \frac{\frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})^4}{\left[\frac{1}{n-1} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})^2 \right]^2}.$$

It can be shown that for a standard normal distribution, the kurtosis is approximately 3. So, instead of using the kurtosis, we use the excess kurtosis, which is just the kurtosis minus 3. In this way, if the excess kurtosis

- is positive, we know right away that the tails of the distribution are heavier than for a standard normal;
- is negative, then the distribution has thinner tails than a normal distribution.

There are distributions that exhibit kurtosis. One example would be the Student's t-distribution. The Student's t-distribution is symmetric (but also comes in an asymmetric version). Unlike the normal distribution, the Student's t has excess kurtosis. Instead of being described by a mean and standard deviation, the Student's t-distribution is described by a single number: the degrees of freedom. When the number of degrees of freedom is relatively small, for example 4, then there is indeed a high amount of excess kurtosis in the distribution. As the number of degrees of freedom increases, the distribution asymptotically approaches the normal.

For now, let's think of having skewness or excess kurtosis as discrediting the use of the normal distribution to explain the data. We will say lots about both, particularly kurtosis, throughout this program as we examine tails and risk management and portfolio problems.

The bottom line is that if returns are normal, our jobs are much easier. However, it is not really up to us to decide that but rather to assess it. We must examine the data and shake it enough to see if normality still holds. If returns turn out to be normally distributed, then our job is much easier. This is because the mean and standard deviation will be proper representatives for making estimations of percentiles and probabilities. If, on the other hand, the normal distribution is inappropriate, then we can merely switch to a distribution that reflects the skewness, kurtosis, or the combined skewness and kurtosis. Such a distribution may involve some additional skills in modeling. We may also decide to use no distribution at all but rather keep the original data and use what is known as a nonparametric method.

Students can utilize [this Shiny application](#) to visually assess the normality of various securities. It is recommended to select date ranges that encompass periods of high volatility. This approach will allow you to observe how asset distributions change during these volatile periods, providing insight into the behavior of financial instruments under different market conditions.

For this module, the emphasis has been on volatility. What did we learn about volatility? From a financial point of view, volatility represents the amount of fluctuation investors are willing to tolerate. Some investors are willing to take more risks than others. Those investors essentially are receptive to having amounts of volatility. Other investors may shun the idea of excessive risk. For these investors, too much volatility is unsettling. The choice of volatility is the investor's preference. Our job as financial engineers is to find metrics and models that can represent volatility, whether it is dynamic in nature, is in violation of simple models, or requires advanced ways of thinking.

7. Conclusion

This is the final lesson of the module. When discussing single securities, volatility is one of the most important statistical properties, if not the most important property, that a security has. The problem only gets more difficult when we consider securities in pairs. What will be very important is to understand how two or more securities work together. For this, we will turn to another statistical challenge in the financial world, namely correlation. We will revisit all the topics of stocks and cryptocurrencies in the next lesson but will add to our toolkit the important notion of correlation.

References

- Netflix, Inc (NFLX). *Yahoo Finance*. <https://finance.yahoo.com/chart/NFLX/>. Accessed 28 Feb. 2022.